

Supplementary Figures S1-S10.

Phylogenetic tree of 54 species of Salamandridae with ten individual articulations (1 - 10) mapped as continuous traits. Tip colours range from red to blue corresponding to range 0 to 1 (from entirely fused to entirely separated bones). Maximum-likelihood ancestral-state reconstruction under the best-supported Ornstein–Uhlenbeck model was estimated at the root of the tree, representing the inferred cranial kinesis of the most recent common ancestor of all sampled taxa. Abbreviations: SMR – Salamandrininae. PLE – Pleurodelinae and SAL – Salamandrinae.

1. Supplementary Figure S1. Character state reconstruction for the premaxilla–maxilla articulation.
2. Supplementary Figure S2. Character state reconstruction for the nasal–frontal articulation.
3. Supplementary Figure S3. Character state reconstruction for the prefrontal–frontal articulation.
4. Supplementary Figure S4. Character state reconstruction for the frontal–parietal articulation.
5. Supplementary Figure S5. Character state reconstruction for the midline contact between the left and right frontal and parietal bones.
6. Supplementary Figure S6. Character state reconstruction for the parietal–otico-occipital articulation.
7. Supplementary Figure S7. Character state reconstruction for the orbitosphenoid – skull roof and parasphenoid articulations.
8. Supplementary Figure S8. Character state reconstruction for the vomer–parasphenoid articulation.
9. Supplementary Figure S9. Character state reconstruction for the parasphenoid–otico-occipital articulation.
10. Supplementary Figure S10. Character state reconstruction for the squamosal–otico-occipital articulation.

Supplementary Figures S11.

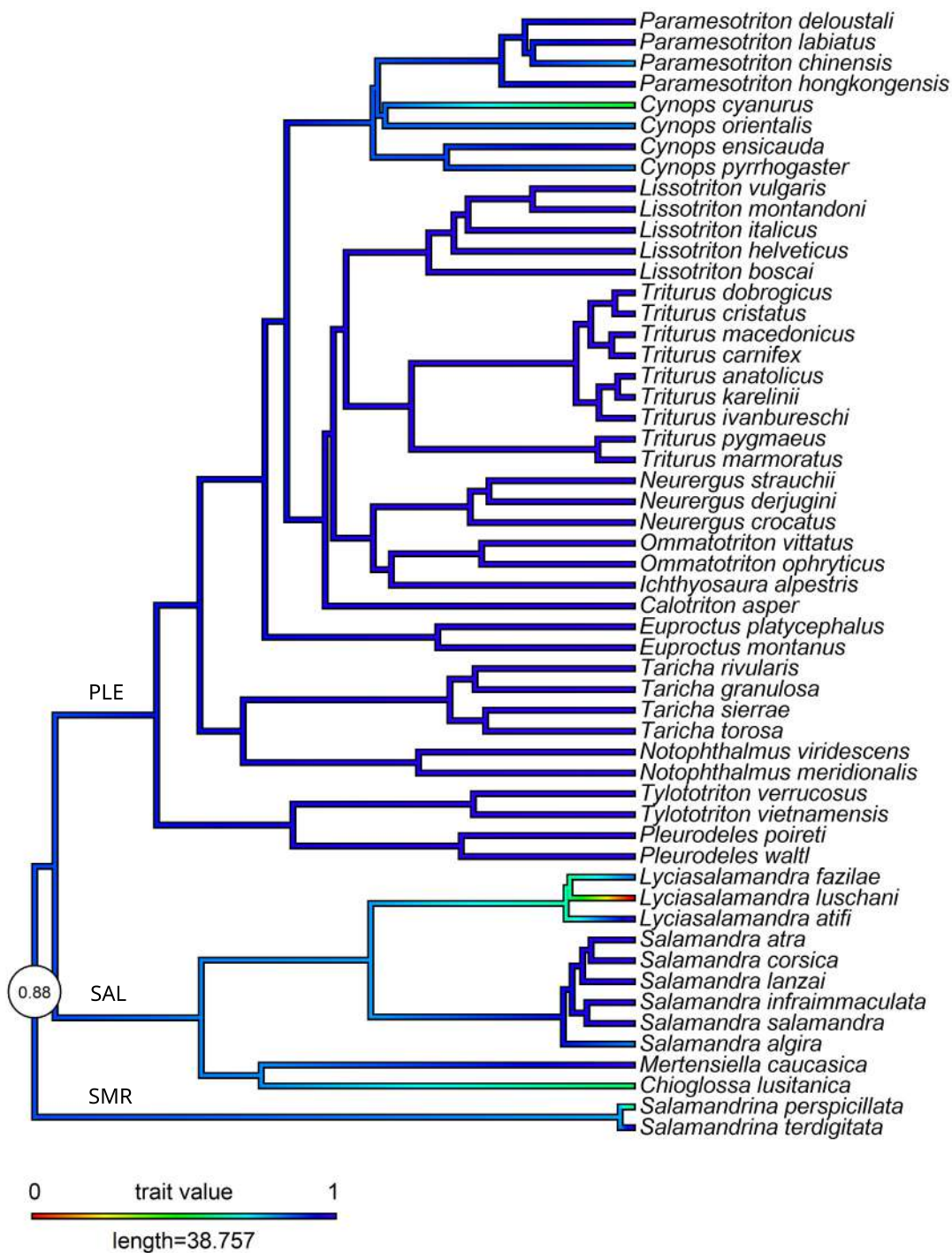
Phylogenetic tree of 54 species of Salamandridae with skull size (CS in mm) mapped as a continuous trait. Tip colours range from red to blue corresponding to range from CS $\approx$ 16.5 to CS $\approx$ 46.2. Maximum-likelihood ancestral-state reconstruction under the best-supported Ornstein–Uhlenbeck model was estimated at the root of the tree, representing the inferred skull size of the most recent common ancestor of all sampled taxa. Abbreviations: SMR – Salamandrininae. PLE – Pleurodelinae and SAL – Salamandrinae.

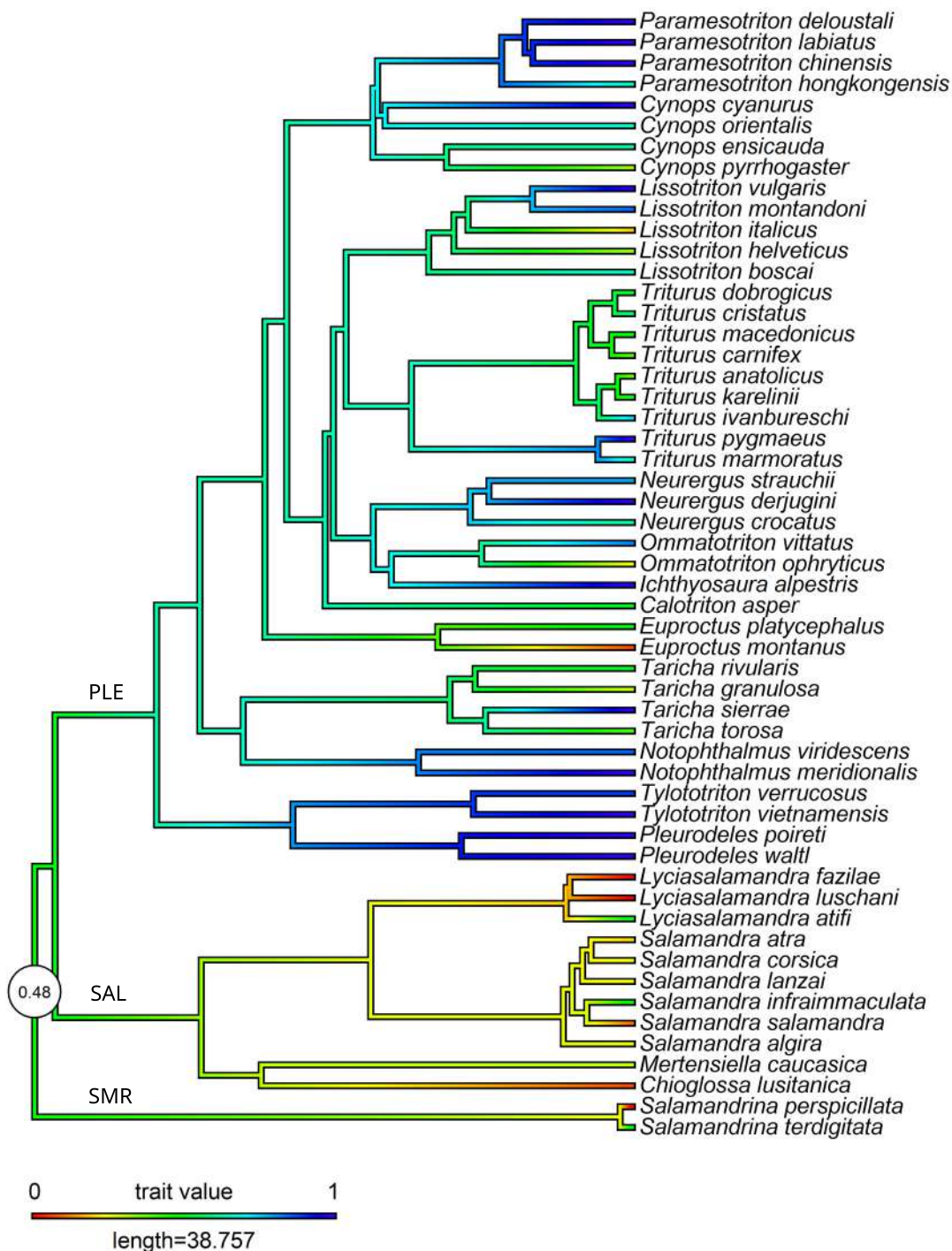
Supplementary Figures S12.

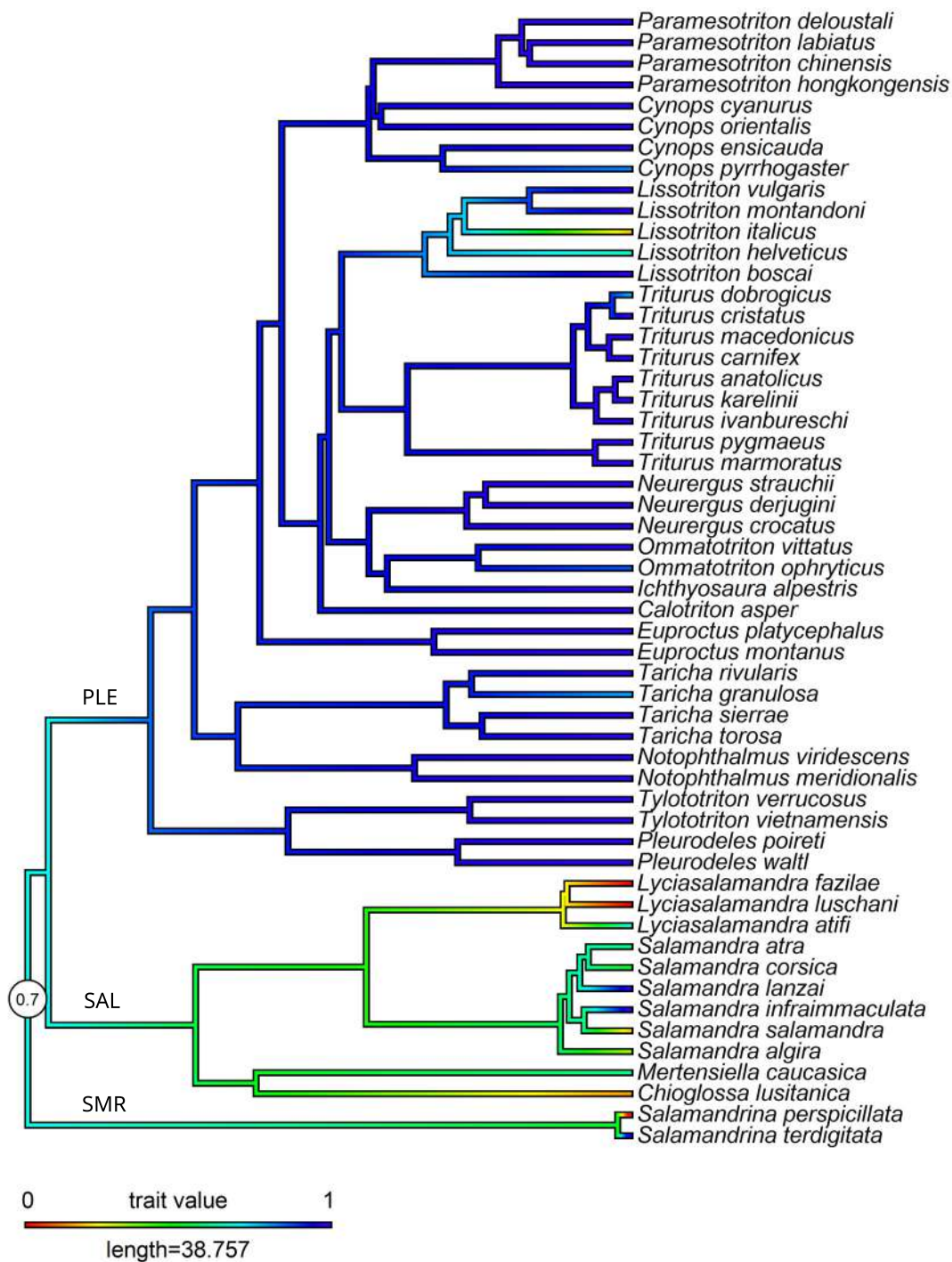
Phylogenetic tree of 54 species of Salamandridae with corrected bone thickness (skull robustness, SR) mapped as a continuous trait. Tip colours range from red with SR = 0.55 for thin bones to blue for SR = 2.77 for thick bones. Maximum-likelihood ancestral-state reconstruction under the best-supported Ornstein–Uhlenbeck model estimated SR $\approx$ 1.15 at the root of the tree, representing the inferred skull robustness of the most recent common ancestor of all sampled taxa. Abbreviations: SMR – Salamandrininae. PLE – Pleurodelinae and SAL – Salamandrinae.

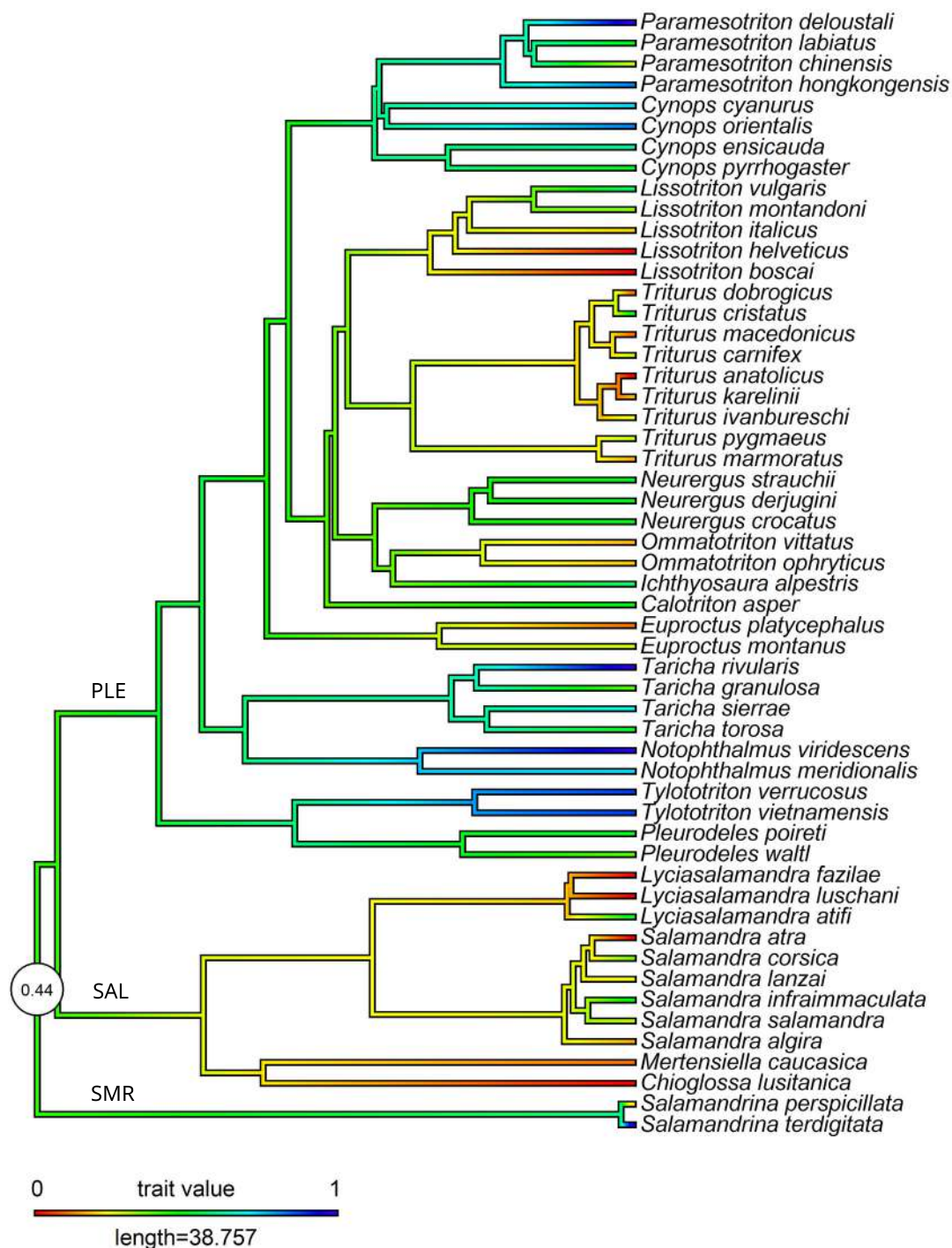
Supplementary Figures S13.

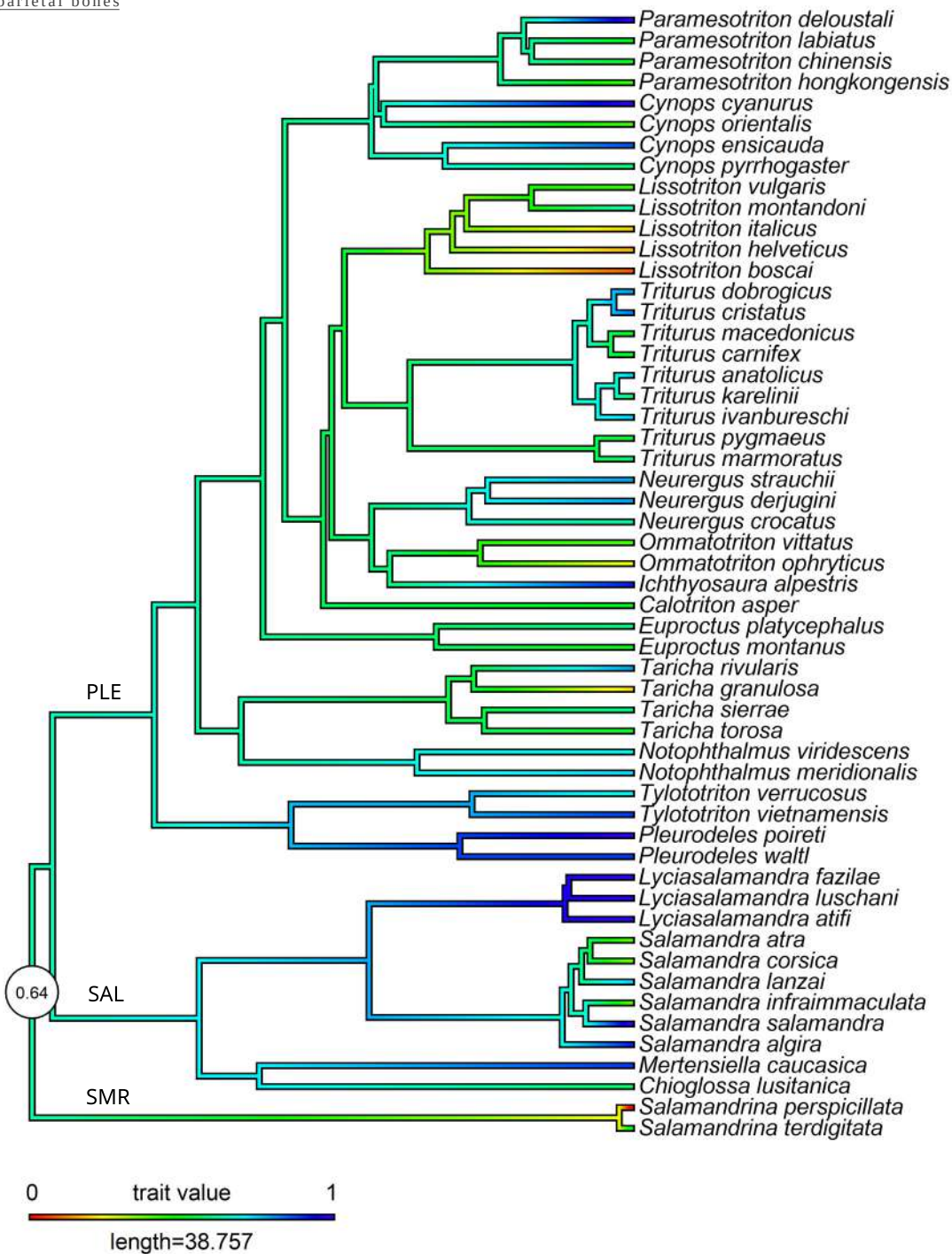
Phylogenetic tree of 54 species of Salamandridae with species ecological preferences (duration of the aquatic period) mapped as a continuous trait. Tip colours range from red to blue corresponding to range from terrestrial (character state 1) to aquatic (character state 0). Maximum-likelihood ancestral-state reconstruction was estimated at the root of the tree, representing the inferred ecological preferences of the most recent common ancestor of all sampled taxa. Abbreviations: SMR – Salamandrininae. PLE – Pleurodelinae and SAL – Salamandrinae.

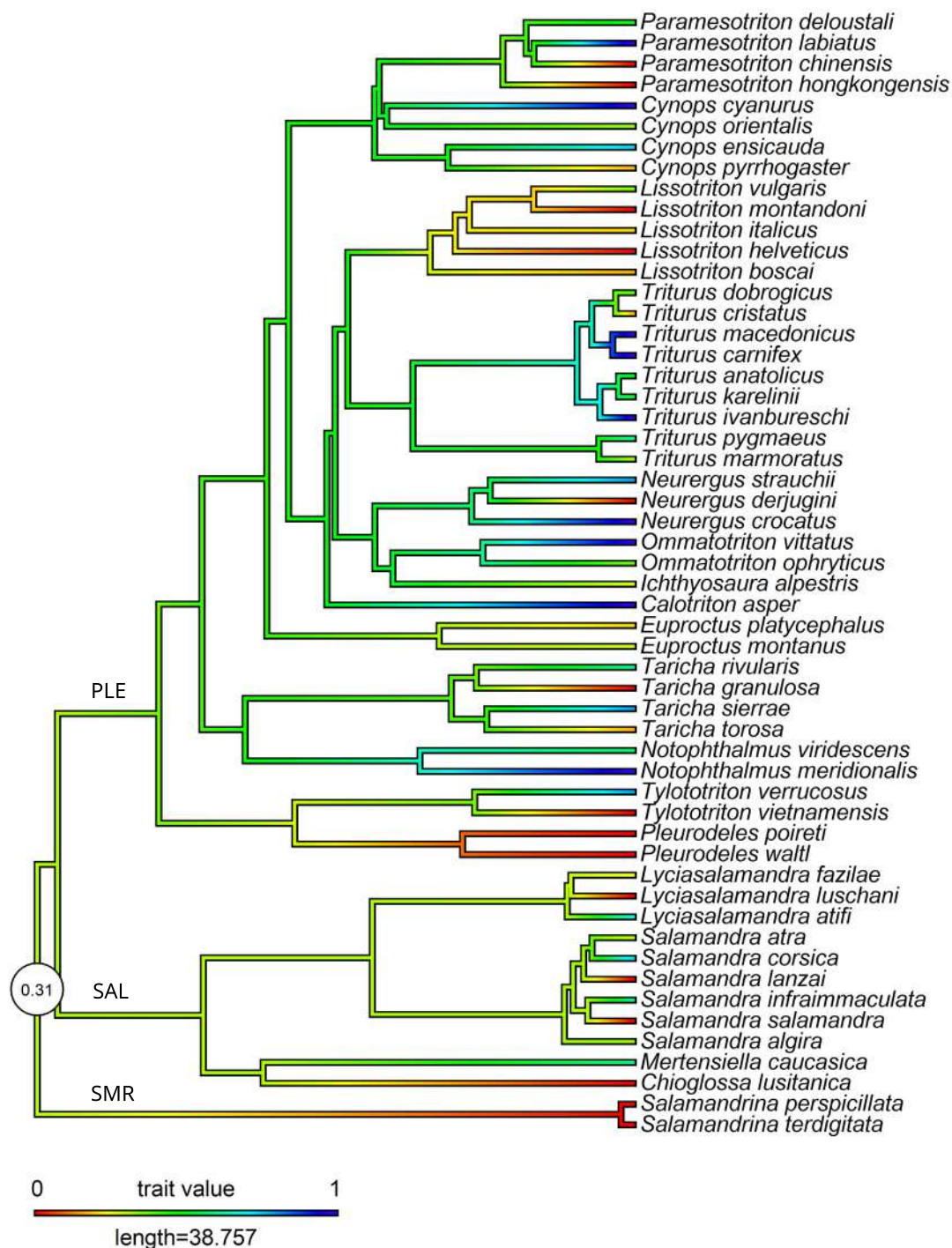


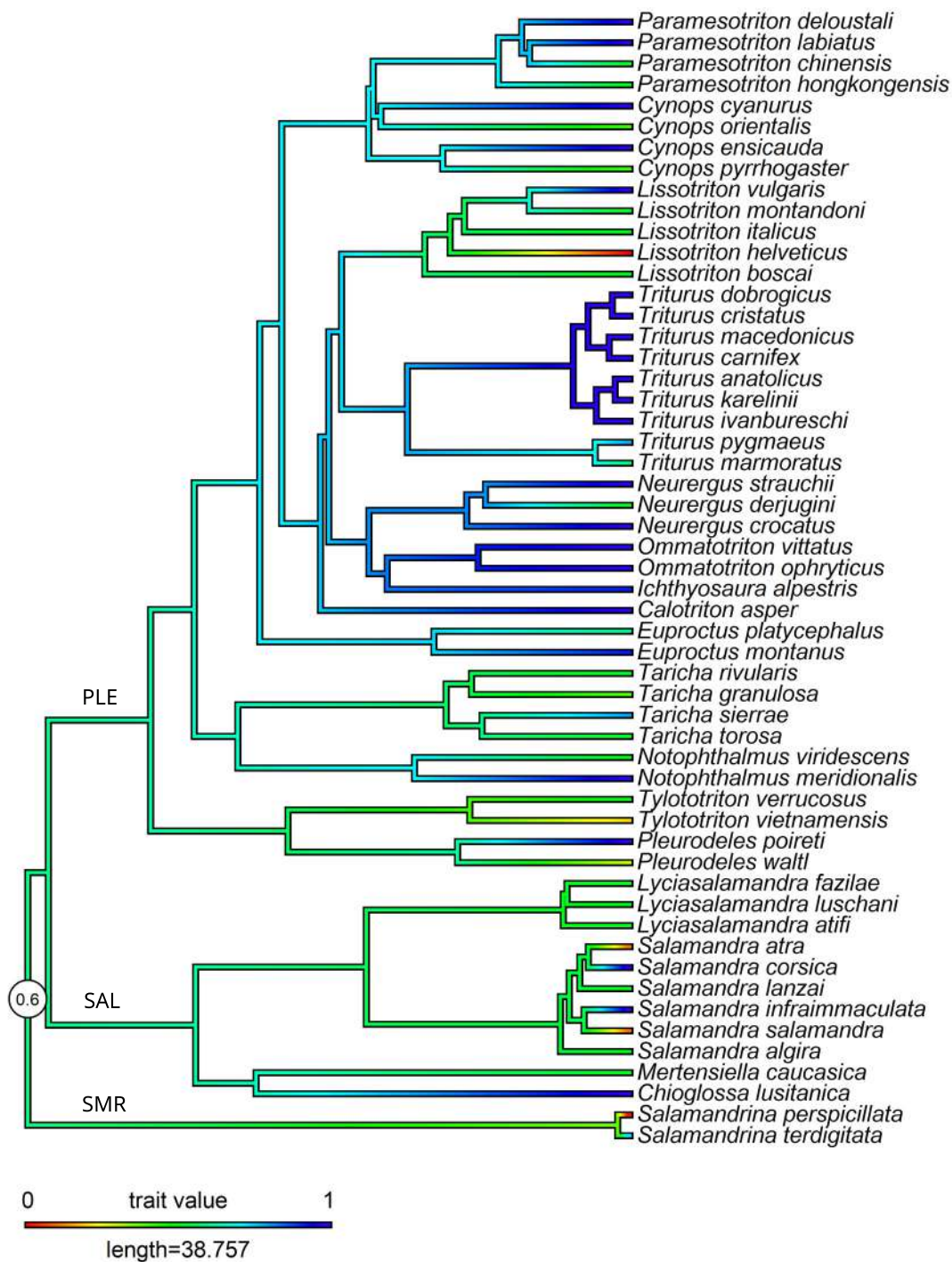


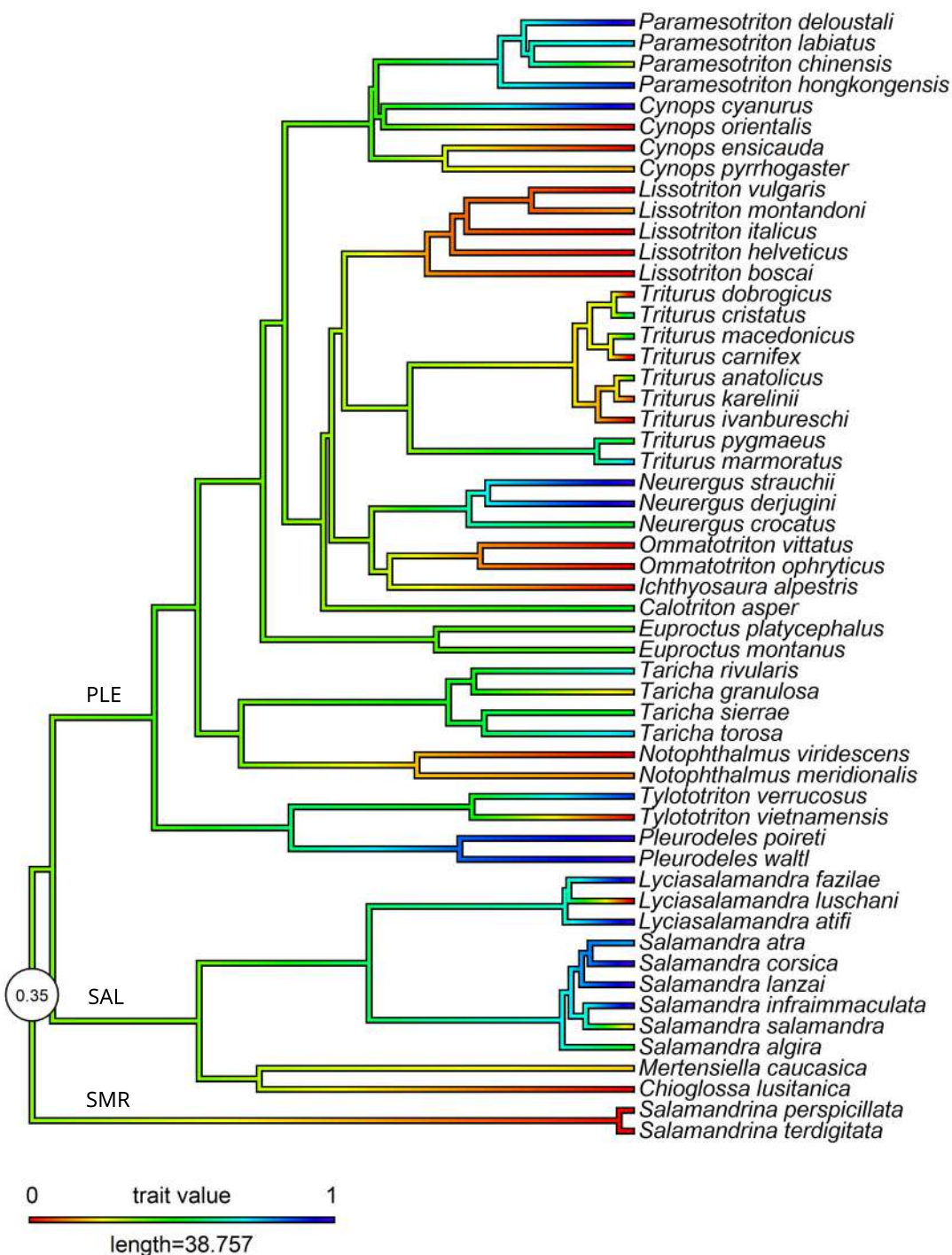




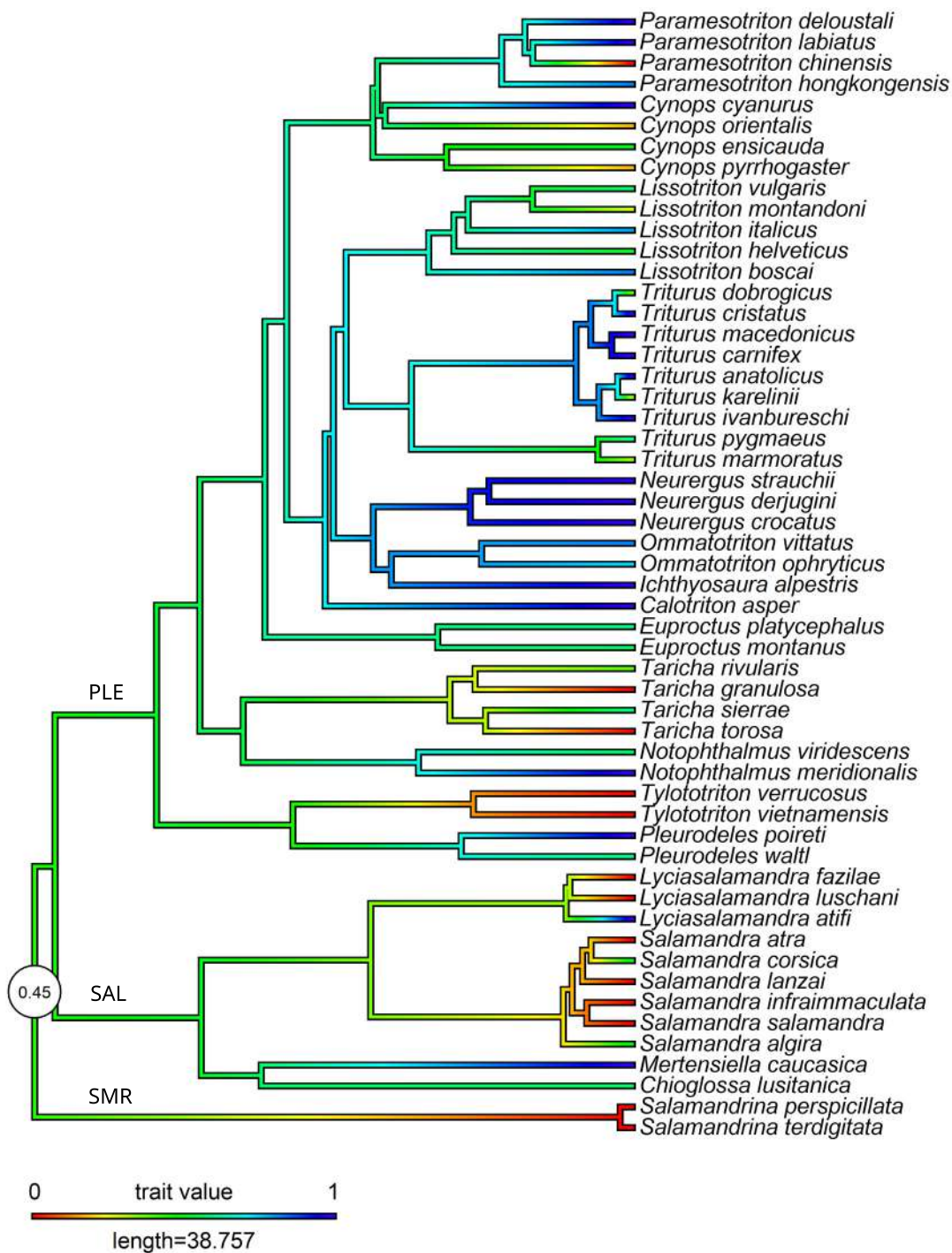


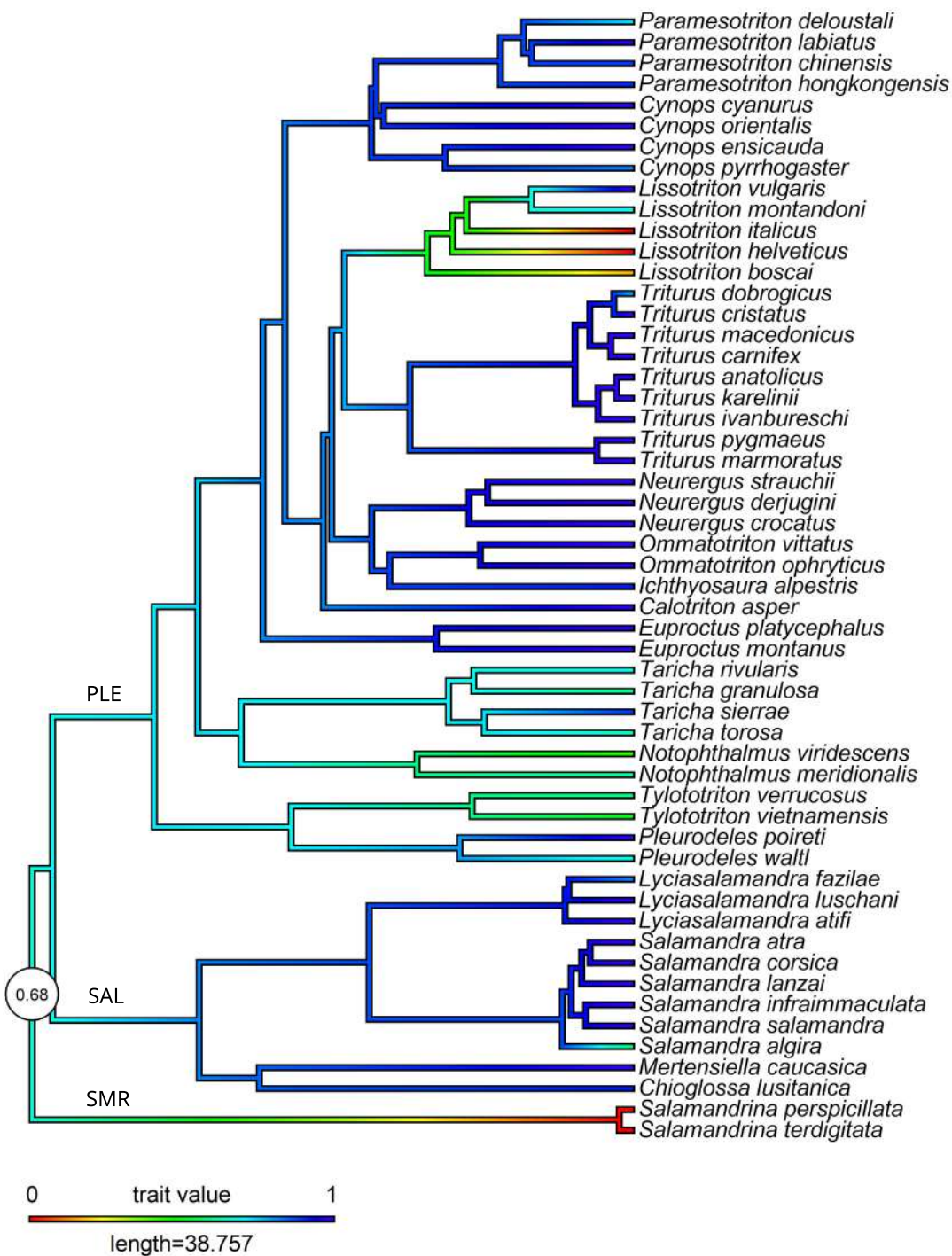


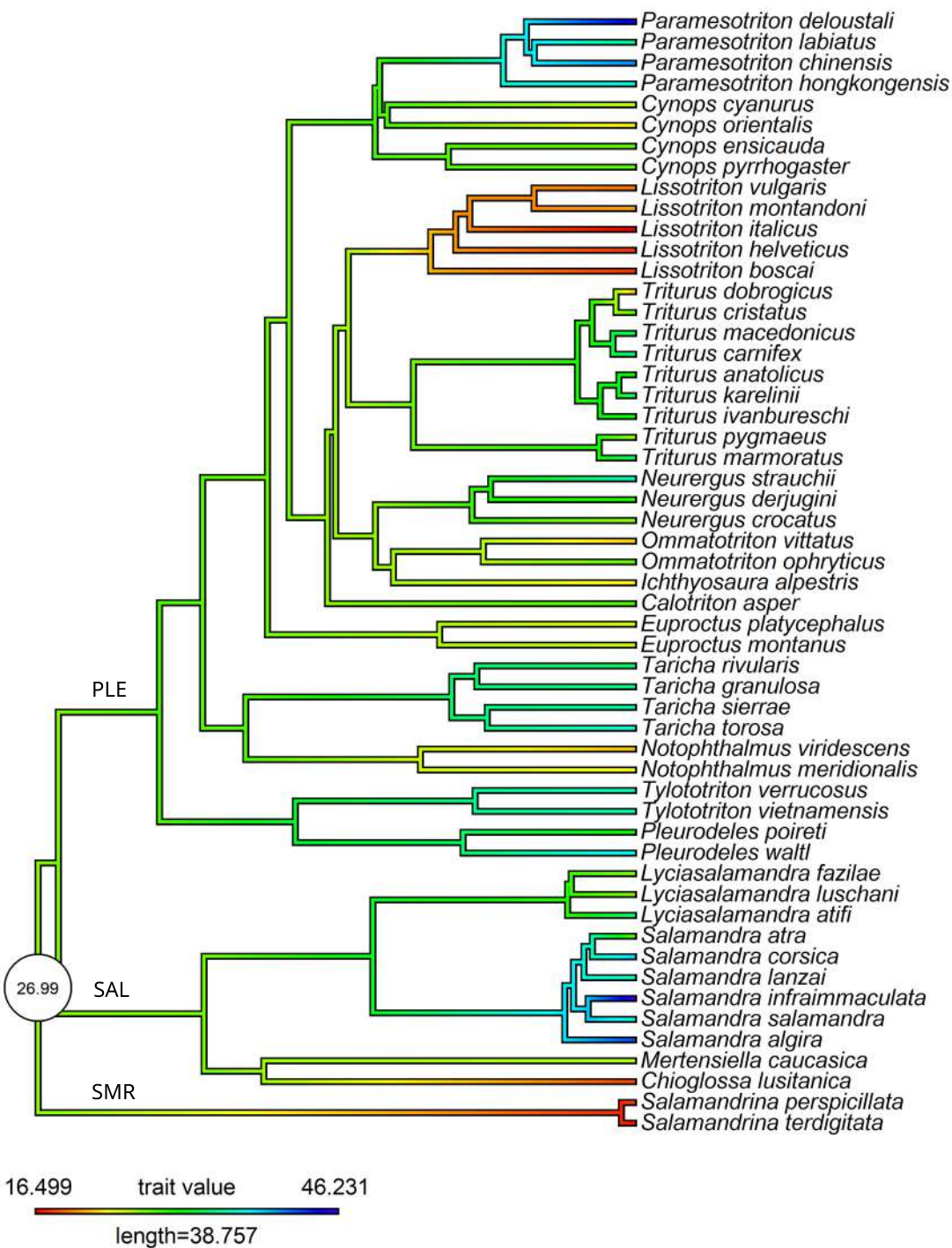




Supp.Fig. S9. Character state reconstruction for the parasphenoid–otico-occipital articulation







Supp.Fig. S12. Character state reconstruction for the bone thickness (skull robustness)

